

ANDREW NOONAN

[Website](#) ◇ [LinkedIn](#) ◇ andrew.p.noonan@vanderbilt.edu ◇ 203-635-3000

EDUCATION

Vanderbilt University

Master of Science in Mechanical Engineering

GPA: 4.0/4.0

Nashville, TN

August 2025

Vanderbilt University

Bachelor of Engineering in Mechanical Engineering

Minor in Computer Science | GPA: 3.83/4.0 | Cum Laude

Nashville, TN

May 2023

WORK EXPERIENCE

Vanderbilt Aerospace Design Laboratory (VADL)

Graduate and Undergraduate Research

Nashville, TN

May 2022 - Present

- Designed, fabricated, and characterized performance of Venturi air injectors for viscous fluids
- Modified VW Diesel engine test facility to measure lubricant flow rate and main gallery pressure
- Studied applicability of Fiber Bragg Gratings to structural health monitoring of composite airframes
- Designed, tested, and flight-validated RF payload detachment system for NASA USLI challenge

Vanderbilt University School of Engineering

Teaching Assistant - Energetics Lab, FEA, Fluid Mechanics, Machining Lab

Nashville, TN

August 2022 - December 2024

- Energetics Lab - Guided students through hands-on studies of Fluid Mechanics, Heat Transfer, and Thermodynamics with SI engine, wind tunnel, water tunnel, and heat exchanger labs
- FEA - CNC Machined and tested students' final projects, lectured, and guided students through assignments

SitelogIQ

Solutions Development Intern

Nashville, TN

May 2021 - August 2021

- Created customer-facing technical documents and project reports
- Wrote Python and Power Automate scripts to automate tasks and facilitate communication between teams

LEADERSHIP AND SERVICE EXPERIENCE

NASA University Student Launch Initiative (USLI)

Team President, Graduate Mentor

Nashville, TN

August 2022 - Present

- President (2022–2023): Led a 10-person team in the design, fabrication, and launch of a high-powered rocket featuring a novel robotic payload for NASA's USLI competition; 3rd place nationally among 50+ teams.
- First place in STEM outreach and altitude accuracy awards, led hands-on STEM educational activities to 1400+ K-12 students and community members
- Graduate mentor to 2024 and 2025 teams, outreach lead for 2024 team

AIAA Reusable Launch and Space Vehicle Technical Committee

Awards Subcommittee

Nashville, TN

December 2024 - Present

- Working with academia and industry professionals to develop a student competition for the AIAA RLSV technical committee

AWARDS AND HONORS

- **Leadership Awards:** Vanderbilt Mechanical Engineering Department Leadership Award and Connecticut Associate of Boards of Education for undergraduate and high school leadership respectively
- **Best Paper Award:** Second author, recipient of best paper award at AIAA Student Conference Region II

SKILLS AND INTERESTS

- **Skills & Programming:** Fluid systems, CAD/CAM, MATLAB/Simulink, LabVIEW, C++, Ansys, Comsol
- **Interests:** Travel, live music, cooking, basketball, golf, stand-up comedy, running